AI-Enabled-Intelligent-Study-Assistant-to-Enhance-Student-Learning-Outcomes

INDIVIDUAL THESIS

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Individual Component: Hallucination- Controlled LLM-Based Tutor

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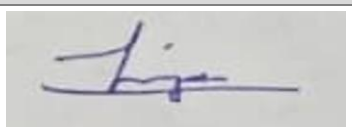
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ABSTRACT

Large Language Models (LLMs) offer transformative potential for personalized and scalable education, yet their well-documented tendency to "hallucinate" generating plausible but factually incorrect or unsupported information presents a serious liability in academic settings. A student who receives a confidently stated but incorrect explanation from an AI tutor may develop persistent misconceptions that are difficult to correct, potentially undermining the very learning process the system is intended to support.

This research presents the design, implementation, and evaluation of the **Hallucination-Controlled LLM-Based Tutor**, the core AI backend component of the "AuraLearn" intelligent learning platform. The system employs a multi-layered hallucination mitigation strategy built around a Retrieval-Augmented Generation (RAG) pipeline. Student queries are semantically matched against a vector database populated with instructor-provided course materials. The retrieved context grounds the Google Gemini LLM's generation, constraining it to curriculum-specific knowledge. Critically, every generated response is then evaluated by a novel **multi-signal Confidence Scorer**, which computes a composite confidence score from three orthogonal signals: (1) retrieval confidence based on cosine similarity of retrieved vector embeddings, (2) a semantic grounding score measuring the alignment between the LLM response and the retrieved context using Sentence Transformer embeddings, and (3) an optional self-consistency score derived from comparing multiple independent LLM generations on the same query with temperature perturbation.

Based on this composite score, a tripartite **Adaptive Responder** module classifies the output as HIGH, MEDIUM, or LOW confidence and formats the response accordingly: HIGH-confidence queries receive a direct, cited answer with source references; MEDIUM-confidence queries receive a pedagogically guided hint with suggested readings that encourages students to engage with source materials; LOW-confidence queries trigger a structured clarification request that explains the system's uncertainty without attempting to answer and redirects the student appropriately. All interactions are logged in full detail to an SQLite database, enabling ongoing research evaluation of hallucination rates, confidence-response alignment, and user satisfaction through a thumbs-up/thumbs-down feedback mechanism.

The backend is implemented in Python using the Fast API framework, containerized with Docker, and exposes a well-documented REST API consumed by the "AuraLearn" Next.js frontend. The individual contribution of IT22120120 encompasses the complete backend architecture design, the multi-signal confidence scoring algorithm, the adaptive response logic, the document ingestion pipeline, the SQLite analytics system, and the Docker deployment configuration.

Keywords: Large Language Models, Hallucination Control, Retrieval-Augmented Generation (RAG), AI Tutor, Educational Technology, Confidence Scoring, Self-Consistency, Semantic Grounding, Fast API, Chroma DB, Sentence Transformers

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CORS	Cross-Origin Resource Sharing
HTTP	Hypertext Transfer Protocol
ITS	Intelligent Tutoring System
JSON	JavaScript Object Notation
LLM	Large Language Model
ML	Machine Learning
NLI	Natural Language Inference
NLP	Natural Language Processing
PDF	Portable Document Format
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer
SLIIT	Sri Lanka Institute of Information Technology
SQLite	SQL Lite (Embedded Relational Database)

1 INTRODUCTION

1.1 Overall Project Context

The “AuraLearn” platform is a research-oriented intelligent educational system developed by a four-member research group at the Sri Lanka Institute of Information Technology as part of the final-year Bachelor's research project (Group ID: R26-SE-022). The platform is designed to augment the traditional university learning management experience with targeted artificial intelligence capabilities that improve student engagement, knowledge retention, and study efficiency.

The platform comprises four independent, containerized microservice components:

1. **Hallucination-Controlled LLM-Based Tutor**: A Fast API backend service that accepts student questions, retrieves relevant course material from a vector database, generates grounded responses via the Google Gemini LLM, computes a composite confidence score to detect potential hallucination, and adaptively formats the response based on that confidence.
2. **AI-Powered Lecture Summarizer**: A backend service that ingests lecture slides and recordings, produces AI-generated structured summaries, identifies learning objectives, and organizes content by weekly topics.
3. **Adaptive Reminder System**: A notification engine that analyses student study patterns and sends personalized revision reminders at optimal intervals.
4. **Frontend Interface**: A Next.js web application that unifies all three backend services into a single student-facing dashboard.

The group's overarching research objective is to investigate how AI can be responsibly integrated into a university learning environment improving educational outcomes while proactively managing the risks associated with AI-generated misinformation.

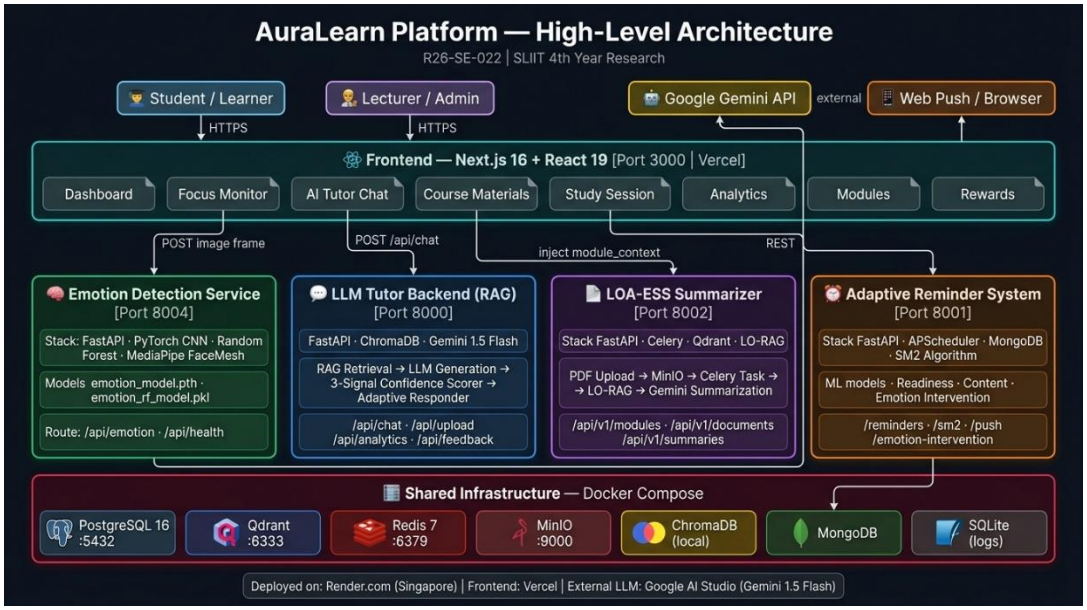


Figure 1: AuraLearn Platform High-Level Architecture

The platform follows a microservices architectural pattern. Each component is independently containerized using Docker and communicates through well-defined REST API contracts. The shared docker-compose.yml file provisions the infrastructure services required across the platform: a PostgreSQL relational database, a Qdrant vector database used by the summarizer, a Redis cache, and MinIO S3-compatible object storage. The LLM tutor backend, however, uses a lightweight local JSON-based vector store and a built-in SQLite database, making it the most self-contained service in the platform.

1.2 Individual Component Context

This thesis documents the **Hallucination-Controlled LLM-Based Tutor** component in full. This component is both the most technically complex and the most research-critical of the four "AuraLearn" services. While the summarizer and reminder components augment the learning experience, the LLM tutor is the primary AI-to-student interaction point and it is therefore where the risk of hallucination is greatest and most consequential.

The component addresses a fundamental tension inherent in deploying LLMs for academic assistance: these models are extraordinarily capable at generating fluent, pedagogically appropriate text, but they do so by predicting statistically likely continuations of text rather than by consulting authoritative sources. When a student asks about a concept covered in their specific lecture slides, a generic LLM may draw on general internet knowledge that contradicts, omits, or misrepresents the specific approach taught in the course. This problem LLM hallucination in an educational context is the research problem this component directly addresses.

The solution is built on four core stages:

Stage	Module	Description
1. Document Ingestion	ingestion/	PDF text extraction → chunking → embedding → vector store
2. Retrieval-Augmented Generation	retrieval/ + llm/llm_engine.py	Semantic retrieval → context injection → Gemini generation
3. Confidence Scoring	llm/confidence_scorer.py	Multi-signal composite confidence computation
4. Adaptive Response	llm/adaptive_responder.py	Confidence-driven response formatting

1.3 Problem Statement

How can an LLM-based tutoring system be designed to provide personalized educational assistance grounded in specific course materials, while actively detecting and mitigating hallucinations, so that students receive either an accurate, cited answer when the system is confident, or appropriate pedagogical redirection when the system's confidence is insufficient without requiring manual review of every response by a human instructor?

1.4 Research Gap

Current AI tutoring systems fall into two broad categories. The first uses raw, general-purpose LLMs without any grounding or confidence evaluation. These systems are capable but produce hallucinations that mislead students and cannot be constrained to a specific curriculum. The second uses RAG pipelines that retrieve relevant documents and inject them into the LLM prompt an improvement, but with no post-generation mechanism to detect whether the LLM actually respected the retrieved context.

The gap this research addresses is the absence of a **post-generation confidence evaluation layer** that simultaneously:

- Measures how semantically aligned the generated response is with the retrieved context (grounding score).
- Measures consistency of the model's answer across multiple independent generations under temperature perturbation (self-consistency).
- Combines these signals with retrieval-level relevance into a principled composite score that drives a pedagogically appropriate adaptive response strategy.

No, widely deployed educational AI system currently implements all three signals in a unified, configurable pipeline with an adaptive output mechanism. This research directly fills that gap.

1.5 Aim and Objectives

Aim: To design, implement, and evaluate a hallucination-controlled LLM-based tutoring backend that quantifies response reliability through a novel multi-signal confidence scoring mechanism and adapts its pedagogical output format based on that confidence.

Objectives:

1. To develop a robust PDF document ingestion pipeline using PyPDF2, NLTK, and Sentence Transformers that converts course materials into a searchable, persistent vector store.
2. To implement a Retrieval-Augmented Generation pipeline using cosine similarity search that contextualizes LLM generation with the most relevant course content.
3. To design and implement a Confidence Scorer module that computes a weighted composite confidence score from retrieval similarity, semantic grounding, and optional self-consistency signals.
4. To implement an Adaptive Responder module that routes responses through three distinct formatting strategies (Direct Answer, Guided Hint, Clarification Request) based on configurable confidence thresholds.
5. To build a comprehensive interaction logging system using SQLite that captures all interactions with full confidence metadata and supports student feedback and aggregate analytics.
6. To expose all capabilities through a well-structured, self-documented Fast API REST API containerized with Docker for reproducible deployment.
7. To integrate with the "AuraLearn" summarizer backend, enabling the tutor to use AI-generated lecture summaries as supplementary RAG context when the local vector store is sparse.

1.6 Scope, Limitations, and Assumptions

In Scope:

- Backend API design and implementation using Python and Fast API.
- PDF ingestion pipeline (text extraction, word-based chunking, batch embedding, JSON vector storage).
- RAG retrieval using cosine similarity over Sentence Transformer (all-MiniLM-L6-v2) embeddings.
- LLM integration with Google Gemini API (model: gemini-2.5-flash).

- Multi-signal confidence scoring (retrieval confidence, semantic grounding, optional self-consistency).
- Adaptive response formatting (three output strategies).
- Interaction logging, student feedback, and aggregate analytics via SQLite.
- Docker containerization and environment variable configuration.
- Integration interface with the "AuraLearn" summarizer backend.

Out of Scope:

- Frontend user interface design and implementation (separate group member).
- Lecture summarization and learning objective generation (summarizer component).
- Adaptive reminder scheduling and delivery (separate group member).
- Fine-tuning or retraining the base LLM model.
- Authentication, user management, and authorization (platform-level concern).
- Production-grade scalability (system is designed for research prototype scale).

Assumptions:

- Uploaded PDF course materials are treated as the authoritative ground truth for the tutoring domain.
- The Google Gemini API is accessible during operation with a valid API key.
- Students interact with the system through the "AuraLearn" frontend; direct API access by end users is not a primary scenario.
- The self-consistency check (three sequential API calls) introduces acceptable latency for research evaluation purposes.

Limitations:

- Confidence scoring is probabilistic; it reduces but cannot eliminate the risk of undetected hallucinations.
- Semantic similarity measures topical alignment rather than logical entailment — a response can score well on grounding while containing subtle factual errors.
- The JSON-file vector store is appropriate for prototype scale but would require replacement with a production vector database for large document collections.
- System quality is bounded by the completeness and quality of uploaded course materials.

1.7 Individual Contribution

The following work represents the exclusive individual contribution of IT22120120 to the "AuraLearn" platform:

Area	Files / Modules	Description
Problem formulation	—	Research gap identification and problem statement
Backend architecture	main.py	Fast API app structure, lifespan, global singletons, all 12 endpoints
Configuration	config/settings.py	Pydantic Settings with 19 env-var parameters
PDF ingestion	ingestion/pdf_loader.py	PDF file discovery
PDF ingestion	ingestion/chunker.py	NLTK tokenization, text cleaning, overlap chunking
PDF ingestion	ingestion/embedder.py	PyPDF2 extraction, Sentence Transformer batch encoding, vector store persistence
PDF ingestion	ingestion/ingest_pipeline.py	Orchestration of full ingestion workflow
Retrieval	retrieval/query_embedder.py	Query string embedding
Retrieval	retrieval/retriever.py	JSON vector store, cosine retrieval, metadata management, delete by filename
LLM integration	llm/llm_engine.py	Google Gemini API client, system prompt engineering, single + multi-response generation
Confidence scoring	llm/confidence_scorer.py	Full 3-signal composite scorer (retrieval, grounding, self-consistency)
Adaptive response	llm/adaptive_responder.py	Three-tier response formatting logic
Logging & analytics	database/query_logger.py	SQLite schema, log, feedback, 15-metric analytics engine
Containerization	Docker file	Python 3.11-slim container, dependencies, port 8002

1.8 Thesis Structure

- **Chapter 1** — Introduces the project context, individual component, problem statement, research gap, aim and objectives, scope, limitations, and individual contribution.
- **Chapter 2** — Reviews literature on LLMs in education, hallucination taxonomy, Retrieval-Augmented Generation, confidence estimation methods, and adaptive response strategies in Intelligent Tutoring Systems.
- **Chapter 3** — Presents the research methodology: design-science approach, research phases, data collection strategy (interaction log database), experimental procedure (four controlled scenarios), and quantitative evaluation metrics.
- **Chapter 4** — Details the product and solution design: functional and non-functional requirements, microservice architecture, API specification, confidence scoring design, adaptive response design, and data persistence design.
- **Chapter 5** — Describes the implementation: technology stack justification, module-by-module implementation walkthrough (ingestion pipeline, retriever, LLM engine, confidence scorer, adaptive responder, query logger), integration with frontend and summarizer backend, and functional test cases.

- **Chapter 6** — Presents research results: confidence distribution analysis, adaptive response alignment, self-consistency impact, and integration validation results.
- **Chapter 7** — Discusses findings, compares with related work, addresses limitations and threats to validity, and evaluates the product's contribution.
- **Chapter 8** — Considers commercialization: market opportunity, target segments, business model options, and IP considerations.
- **Chapter 9** — Concludes the research and outlines seven future research directions.

2 LITERATURE REVIEW

2.1 Large Language Models in Education

The deployment of Large Language Models in educational settings has attracted growing research attention since the public release of GPT-3 (Brown et al., 2020) and subsequent model generations. LLMs have demonstrated strong performance on academically relevant tasks including reading comprehension, question answering, explanation generation, Socratic dialogue, essay feedback, and code tutoring — capabilities directly applicable to intelligent tutoring systems.

However, the fundamental challenge of factual reliability has constrained practical adoption in high-stakes educational contexts. LLMs trained on broad internet corpora prioritize fluency and plausibility over factual accuracy when generating responses at the boundaries of their training data. In educational contexts, this manifests as AI tutors providing elegant but incorrect explanations, fabricating non-existent citations, or presenting information that directly contradicts the specific pedagogical approach used in a given course — even when that approach is provided as context.

The promise of LLM-based tutoring is significant: instant 24/7 availability, unlimited patience, ability to generate multiple alternative explanations, and scalability to large student populations. The hallucination risk, however, means that naive deployment of off-the-shelf LLMs in educational settings is irresponsible. This tension motivates the need for systems like AuraLearn's LLM tutor, which maintain the accessibility benefits of LLMs while actively mitigating their reliability risks.

2.2 The Hallucination Problem

"Hallucination" in LLMs refers broadly to the generation of content that is factually incorrect, ungrounded in provided context, or internally inconsistent. Ji et al. (2023) provide a comprehensive survey of hallucination in natural language generation, identifying it as one of the most significant barriers to production deployment of generative AI systems.

The literature distinguishes two primary hallucination types relevant to this work:

Intrinsic hallucination: The model contradicts information explicitly provided in the input context. In the AuraLearn tutoring scenario, this would manifest as the LLM generating an answer that directly contradicts a passage in the retrieved course material.

Extrinsic hallucination: The model introduces information not present in the input context and not verifiable from the context alone. In the tutoring scenario, this might appear as the LLM elaborating on a concept using general internet knowledge that goes beyond what the course specifically teaches, potentially introducing information inconsistent with the course's approach.

Maynez et al. (2020) further distinguish faithfulness hallucination (responses unfaithful to provided source material) from factual hallucination (responses that contradict world

knowledge). For the AuraLearn system, faithfulness hallucination is the primary concern: the system is provided with specific course material context, and the risk is that the Gemini model generates responses that deviate from that context and draw on its pre-trained general knowledge instead.

2.3 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation was introduced by Lewis et al. (2020) as a method for grounding LLM generation in external knowledge sources at inference time, without requiring model retraining. The core mechanism involves three stages: (1) encoding both the user query and a document corpus into a shared dense embedding space using a bi-encoder model, (2) retrieving the most semantically similar documents using cosine similarity (or approximate nearest-neighbor search), and (3) conditioning LLM generation on the concatenation of the original query and the retrieved documents.

RAG significantly reduces hallucination by providing the model with relevant, verifiable source material at inference time. The model's generation is conditioned on specific passages from authoritative sources, reducing its reliance on potentially stale or incorrect parametric (training-time) knowledge. For the AuraLearn use case, this means the Gemini model is conditioned on specific passages from the uploaded lecture slides rather than generating from general internet knowledge about a topic.

However, RAG alone does not guarantee faithful generation. Empirical studies have shown that LLMs can still ignore retrieved context and generate responses based on parametric knowledge, particularly when: (a) the retrieved documents are partially relevant but do not perfectly match the query, (b) the model has strong prior beliefs about the correct answer that conflict with the retrieved text, or (c) the retrieved context is long and the model's attention mechanism de-prioritizes certain passages. This is the key motivation for adding a post-generation confidence evaluation layer on top of the RAG pipeline.

Table 1: Comparison of Hallucination Mitigation Approaches in LLM Literature (Page ~10)

Approach	Description	Advantages	Limitations	Applicable to AuraLearn?
Prompt engineering only	Instructing the model to be factual and cite sources	Simple; no infrastructure	Unreliable; model can ignore instructions	Partial (used as part of system prompt)
Retrieval-Augmented Generation (RAG)	Ground generation in retrieved documents	Strong reduction in factual errors	Does not detect if LLM ignores context	Yes (core pipeline)
RLHF / Instruction Tuning	Fine-tune model on human preference data	Improved overall quality and honesty	Resource-intensive; not curriculum-specific	No (cannot fine-tune Gemini)
Verbalized self-confidence	Ask LLM to rate its own confidence	Simple to implement	LLMs are poorly calibrated self-evaluators	No (unreliable signal)
NLI-based Grounding	Use NLI classifier to check response faithfulness	Objective, model-agnostic faithfulness signal	Requires separate trained NLI model	Yes (optional ML grounding signal)

Approach	Description	Advantages	Limitations	Applicable to AuraLearn?
Self-Consistency Sampling	Generate multiple responses; measure agreement	Objective uncertainty proxy; catches inconsistency	High latency; N API calls per query	Yes (third confidence signal)
Multi-signal composite scoring (this work)	Combine retrieval, grounding, and self-consistency	Robust; configurable; adaptive output	Probabilistic; semantic similarity ≠ entailment	Yes — core contribution

2.4 Confidence Estimation and Uncertainty Quantification

Confidence estimation in generative AI is substantially more challenging than in classical classification, where model output probabilities can be directly interpreted as confidence. For auto-regressive LLMs, token-level probabilities are not straightforwardly interpretable as response-level confidence estimates.

Several approaches have been proposed:

Verbalized confidence: Prompting the model to state its own confidence level. This approach is simple but unreliable — studies have consistently shown that LLMs are overconfident in their self-assessments, particularly for domain-specific questions at the edge of their training data.

Sampling-based consistency (Wang et al., 2022): Generating multiple outputs for the same prompt and measuring their semantic agreement. High agreement across multiple independent generations suggests robust, confident knowledge; high variance suggests uncertainty. This is the theoretical basis for the self-consistency signal implemented in this research.

Entailment-based grounding: Using a Natural Language Inference (NLI) model to predict whether the generated response is logically entailed by the retrieved context. A high entailment probability indicates the response faithfully follows from the source material; a low probability indicates potential hallucination. The Confidence Scorer optionally uses an NLI classifier from the Hugging Face transformers library for this purpose.

Semantic similarity: Measuring cosine similarity between embeddings of the generated response and retrieved context. This is a computationally lightweight proxy for grounding that does not require a separate NLI model, though it measures topical overlap rather than logical entailment.

Min et al. (2023) demonstrate that atomic factual accuracy of LLM responses can be estimated from fine-grained entailment scoring. This work applies a related principle at a coarser response granularity, trading some precision for computational tractability in a real-time tutoring scenario.

2.5 Adaptive Response Strategies in Intelligent Tutoring Systems

Classical Intelligent Tutoring Systems (ITS) such as Auto Tutor (VanLehn, 2011) and Carnegie Learning's Cognitive Tutor implement adaptive response strategies based on student performance models. These systems adjust explanation difficulty, scaffolding level, and

feedback format based on a dynamically maintained model of student knowledge. However, these approaches are domain-specific (often requiring hundreds of hours of expert knowledge authoring per domain) and do not transfer readily to new curricula.

The adaptive response strategy in this research takes a different axis of adaptation: rather than adapting to the **student's** knowledge level, the system adapts to its **own confidence** in the correctness of its answer. This is a novel application of uncertainty-driven output adaptation that:

- Is completely domain-agnostic (no per-domain authoring required).
- Transfers immediately to any course materials uploaded to the system.
- Implements a conservative pedagogical safety policy: when uncertain, redirect rather than risk misleading.

The three-tier strategy (Direct Answer → Guided Hint → Clarification Request) corresponds to three points on a spectrum from full confidence to acknowledged ignorance. The guided hint format in particular implements a Socratic teaching strategy — providing direction toward the answer rather than the answer itself — which may produce better learning outcomes than a direct answer even when the system could respond correctly.

2.6 Summary and Research Position

The literature review confirms that RAG is the most practically effective grounding approach for domain-specific LLM applications, but that RAG alone is insufficient for high-stakes educational deployment. Confidence estimation through sampling-based self-consistency and entailment-based grounding provides complementary, objective uncertainty signals that do not rely on the model's own self-assessment.

This research's position is to synthesize these established techniques into a unified, deployable pipeline specifically designed for educational AI safety. The key novelty is the combination of three signals into a composite confidence score, directly connected to an adaptive response strategy that shapes the tutor's pedagogical behavior based on its own quantified uncertainty. This combination grounding-aware RAG + multi-signal confidence + adaptive output — has not been reported as a unified, production-quality implementation in the intelligent tutoring systems literature.

3 RESEARCH METHODOLOGY

3.1 Research Design

This research follows a **design-science methodology** (Hevner et al., 2004): the primary research output is a working software artefact (the LLM tutor backend) that instantiates a novel theoretical contribution (the multi-signal confidence scoring and adaptive response framework). The research is evaluated through both technical validation (functional correctness of the pipeline against controlled test scenarios) and empirical measurement (quantitative metrics computed from real interaction logs).

The research was conducted in five sequential phases:

Phase	Duration	Activity	Output
1. Literature Review	Weeks 1–3	Survey LLM hallucination, RAG, ITS, confidence estimation	Research gap, signal selection rationale
2. Architecture Design	Weeks 4–5	System decomposition, API contracts, data schema	Architecture diagrams, API specification
3. Implementation	Weeks 6–12	Python implementation of all backend modules	Functional backend codebase
4. Integration & Testing	Weeks 13–14	Integration with frontend and summarizer; functional tests	End-to-end system validation
5. Evaluation	Weeks 15–16	Collect interaction logs, compute metrics, gather feedback	Quantitative research results

3.2 Data Collection Strategy

The primary data source for system evaluation is the interaction log database (data/auralearn_logs.db), populated during real and controlled system operation. Every query-response cycle is automatically persisted to the query logs SQLite table with the following research-relevant fields:

- Student's original query text.
- LLM-generated response text.
- Response type classification (direct answer, guided hint, clarification request).
- Full confidence breakdown: composite score, retrieval confidence, grounding score, optional self-consistency score.
- LLM model used and self-consistency mode active flag.
- Student feedback rating (thumbs up/thumbs down, submitted voluntarily).

Secondary data is collected from four structured test scenarios with controlled inputs and known expected outputs, used to validate the pipeline's functional correctness before empirical evaluation.

3.3 Experimental Procedure

The core evaluation tests the confidence scoring system against four controlled scenarios designed to span the expected range of confidence levels:

Scenario 1 — High-relevance query (expected: HIGH, direct answer): Submit a question whose exact answer is contained verbatim in an uploaded lecture PDF. Expected outcome: high retrieval similarity (>0.75), high grounding score (response closely mirrors retrieved text), composite score ≥0.75, classified as HIGH, formatted as Direct Answer.

Scenario 2 — Partial-relevance query (expected: MEDIUM, guided hint): Submit a question that is topically related to uploaded materials but requires generalization or synthesis beyond any single retrieved chunk. Expected outcome: moderate retrieval similarity (0.45–0.75), moderate grounding, composite score in 0.45–0.75 range, classified as MEDIUM, formatted as Guided Hint.

Scenario 3 — Off-topic query (expected: LOW, clarification request): Submit a question completely outside the domain of uploaded materials (e.g., stock market prices when materials cover software engineering). Expected outcome: low/zero retrieval similarity, empty context, grounding score near zero, composite score <0.45, classified as LOW, formatted as Clarification Request.

Scenario 4 — Self-consistency with high-relevance query (expected: HIGH with third signal): Repeat Scenario 1 with enable_self_consistency: true. Expected outcome: three responses generated with temperature perturbation (0.7, 0.85, 1.0); high pairwise cosine similarity among responses (>0.80); composite score computed with 3-signal weights; response should remain classified as HIGH with third signal included.

3.4 Evaluation Metrics

All metrics are computed from the QueryLogger.get_analytics () method, which executes SQL aggregation queries over the full query_logs table:

Table 8: Research Evaluation Metrics Definition

Metric	Formula	Target
Hallucination Rate (%)	(LOW_count / total) × 100	<25% for relevant material queries
Average Composite Confidence	AVG (confidence score)	≥0.60 for well-covered material
Average Grounding Score	AVG (grounding score)	Correlation with user satisfaction
Confidence-Response Alignment (%)	(aligned count / total) × 100	100% (deterministic by design)
User Satisfaction Rate (%)	(thumbs up / total rated) × 100	Target: ≥70%
False Positive Rate (%)	(HIGH+thumbs_down / HIGH rated) × 100	Target: <15%

4 PRODUCT AND SOLUTION DESIGN

4.1 Functional Requirements

ID	Requirement	Priority
FR-01	Accept PDF course materials via REST API upload and ingest them into a persistent vector store	High
FR-02	Retrieve top-K semantically similar document chunks for a given query using cosine similarity	High
FR-03	Generate a grounded response using the Google Gemini API conditioned on retrieved context	High
FR-04	Compute retrieval confidence from the similarity scores of retrieved documents	High
FR-05	Compute a semantic grounding score from embeddings of the response and retrieved texts	High
FR-06	Optionally compute a self-consistency score from an independent response generation	Medium
FR-07	Compute a composite confidence score as a configurable weighted combination of active signals	High
FR-08	Classify the composite score as HIGH, MEDIUM, or LOW using configurable thresholds	High
FR-09	Format the response as a Direct Answer (HIGH), Guided Hint (MEDIUM), or Clarification Request (LOW)	High
FR-10	Log every interaction to SQLite with full confidence metadata and source information	High
FR-11	Accept student thumbs-up / thumbs-down feedback on individual logged responses	Medium
FR-12	Expose an analytics endpoint computing 15+ aggregate research metrics from the log database	Medium
FR-13	Accept module context from the summarizer backend as a supplementary RAG source	Medium
FR-14	Support multi-turn conversation via a conversation history parameter	Medium
FR-15	Support deletion of individual ingested documents from the vector store	Low
FR-16	Expose raw retrieval results without LLM generation for debugging purposes	Low

4.2 Non-Functional Requirements

ID	Requirement
NFR-01	Performance: Respond to a chat query within 30 seconds under normal conditions (excluding self-consistency mode)
NFR-02	Reliability: Logging failures must not interrupt the chat response flow (non-fatal error handling)
NFR-03	Scalability: Containerized with Docker to support deployment and potential horizontal scaling
NFR-04	Configurability: All thresholds and model parameters configurable via environment variables without code changes

ID	Requirement
NFR-05	Interoperability: Implement CORS headers to support frontend communication from localhost:3000
NFR-06	Maintainability: All modules independently importable, testable, and replaceable
NFR-07	Transparency: Provide complete confidence breakdown in every chat response for frontend display
NFR-08	Portability: Run on Python 3.11 with requirements limited to libraries specified in requirements.txt

4.3 System Architecture

The llm-tutor-backend is a self-contained Python microservice with the following internal module structure:

```
llm-tutor-backend/
├── main.py                      # Fast API app, 12 endpoints, lifespan, global
singletons
├── config/
│   └── settings.py              # Pydantic Settings — 19 environment variable
parameters
├── ingestion/
│   ├── __init__.py
│   ├── pdf_loader.py            # PDF file discovery utility
│   ├── chunker.py               # NLTK tokenization, text cleaning, overlap
chunking
│   ├── embedder.py              # PyPDF2 extraction + Sentence Transformer
encoding + persistence
│   └── ingest_pipeline.py       # Ingestion Pipeline orchestrator class +
convenience functions
├── retrieval/
│   ├── __init__.py
│   ├── query_embedder.py        # Query Embedder — encodes query strings to 384-
dim vectors
│   └── retriever.py             # Retriever — JSON vector store with cosine
retrieval + CRUD
├── llm/
│   ├── __init__.py
│   ├── llm_engine.py            # LLMEngine — Gemini API client, prompt
construction, N-sampling
│   ├── confidence_scorer.py     # ConfidenceScorer — 3-signal composite scoring
algorithm
│   └── adaptive_responder.py    # AdaptiveResponder — 3-tier response formatting
├── database/
│   ├── __init__.py
│   └── query_logger.py          # QueryLogger — SQLite logger + feedback +
analytics engine
├── dlls/                        # Windows-specific SQLite DLL override for
ChromaDB compat
├── Dockerfile                   # Python 3.11-slim, port 8002, uvicorn startup
├── requirements.txt             # 14 Python package dependencies
└── .env.example                 # Template for required environment variables.
```

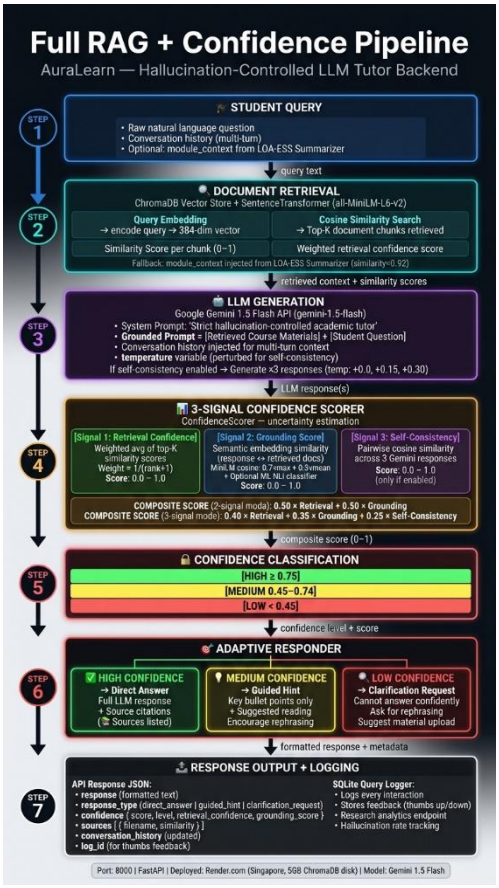


Figure 2: Full RAG + Confidence Pipeline Data Flow

The application lifespan is managed by an async context manager that logs startup status at launch, including the configured vector store path, LLM model name, confidence thresholds, and Query Logger initialization. Global component instances (LLMEngine, ConfidenceScorer, AdaptiveResponder) use a lazy-initialisation singleton pattern via module-level _instance variables and getter functions, ensuring components are created only once per server process and reused across requests.

4.4 API Endpoint Specification

The backend exposes a REST API over HTTP. All endpoints accept and return JSON. The API is auto-documented by FastAPI and accessible at http://localhost:8002/docs (Swagger UI) during development.

Table 3: API Endpoint Specification (Page ~19)

Method	Endpoint	Description	Key Request Fields	Key Response Fields
GET	/health	Service health check	—	{status, version}
GET	/	API directory listing	—	{name, version, endpoints}
GET	/api/llm/status	Gemini API connectivity check	—	{status, provider, model_available, message}

Method	Endpoint	Description	Key Request Fields	Key Response Fields
POST	/api/chat	Main tutoring endpoint — full pipeline	query, conversation_history, top_k, enable_self_consistency, module_context	response, response_type, response_label, confidence, sources, log_id
POST	/api/upload	Upload + ingest a PDF file	Multipart file	{message, filename, documents_count, status}
GET	/api/materials	List ingested document stats	—	{total_chunks, unique_documents, documents, status}
DELETE	/api/materials/{filename}	Remove a document from vector store	Path: filename	{status, message, deleted_chunks, file_deleted}
POST	/api/ingest	Batch ingest from server folder	{pdf_folder}	{message, documents_count, status}
POST	/api/retrieve	Raw vector retrieval (no LLM)	{query, top_k}	{query, results, count}
POST	/api/query	Alias for /api/retrieve	Same	Same
GET	/api/analytics	Aggregate research metrics	—	Full analytics dict (15+ metrics)
GET	/api/logs	Paginated log entries	limit, offset	{logs, total, limit, offset}
POST	/api/feedback	Submit student rating	{log_id, feedback}	{status, log_id, feedback}

Key Request/Response Model Definitions:

```
python
# ChatRequest — the main tutoring request
class ChatRequest(BaseModel):
    query: str                                          # Student's question
(required)
    conversation_history: List[ConversationMessage] = []  # Prior dialogue turns
    top_k: int = 5                                      # Vector retrieval depth
    enable_self_consistency: Optional[bool] = None   # Per-request override for 3-
signal mode
    module_context: Optional[str] = None             # Summarizer-injected lecture
summaries
    module_name: Optional[str] = None                # Module label for source
citation
    module_code: Optional[str] = None                # Module code for source
citation
# ChatResponse — the adaptive tutor response
class ChatResponse(BaseModel):
    response: str             # Formatted adaptive response (Direct Answer / Hint
/ Clarification)
    response_type: str        # "direct_answer" | "guided_hint" |
"clarification_request"
    response_label: str       # "☑ Direct Answer" | "💡 Guided Hint" | "🔍
Clarification Needed"
    confidence: dict          # {score, level, retrieval_confidence,
grounding_score, [self_consistency_score]}
    sources: List[dict]       # [{filename, content[:300], similarity}]
    conversation_history: List[dict]  # Updated history for next turn
    # log_id also returned (not in model, added dynamically)
```

4.5 Confidence Scoring Design

The Confidence Scorer in llm/confidence_scorer.py computes a three-signal weighted ensemble confidence score. The weighting scheme differs based on whether self-consistency is enabled (controlled by the ENABLE_SELF_CONSISTENCY environment variable or the per-request enable_self_consistency flag):

Table 4: Confidence Scoring Signal Weights (Page ~21)

Mode	Signal	Weight	How Computed
Fast (2-signal)	Retrieval Confidence	0.50	Rank-weighted average of cosine similarity scores from vector store
Fast (2-signal)	Grounding Score	0.50	Cosine similarity (response embedding vs. retrieved text embeddings) ± NLI ML model
Full (3-signal)	Retrieval Confidence	0.40	Same as above
Full (3-signal)	Grounding Score	0.35	Same as above
Full (3-signal)	Self-Consistency	0.25	Mean pairwise cosine similarity across N=3 response embeddings

Retrieval Confidence Formula:

```
weights = [1/(i+1) for i in range(n_results)] # rank-based weights: [1.0, 0.5,
0.333, ...]
retrieval_confidence = sum(similarity_i × weight_i) / sum(weights)
```

Grounding Score Formula:

```
# Semantic component (always computed):
response_emb = SentenceTransformer.encode(response_text)
doc_embs = SentenceTransformer.encode(retrieved_texts)
similarities = cosine_similarity(response_emb, each doc_emb)
semantic_grounding = 0.7 × max(similarities) + 0.3 × mean(similarities)
# ML component (if NLI model available):
ml_grounding = NLI_pipeline(claim=response, evidence=joined_context)
# Ensemble:
grounding = 0.5 × ml_grounding + 0.5 × semantic_grounding  # if ML available
grounding = semantic_grounding                               # if ML unavailable
```

Self-Consistency Formula:

```
embeddings = SentenceTransformer.encode(all_N_responses)
pairwise_sims = [cosine(emb_i, emb_j) for all i < j]
self_consistency = mean(pairwise_sims)
```

Confidence Classification Thresholds:

Level	Condition	Response Strategy
HIGH	Composite ≥ 0.75	Direct Answer with citations
MEDIUM	0.45 ≤ Composite < 0.75	Guided Hint with readings
LOW	Composite < 0.45	Clarification Request

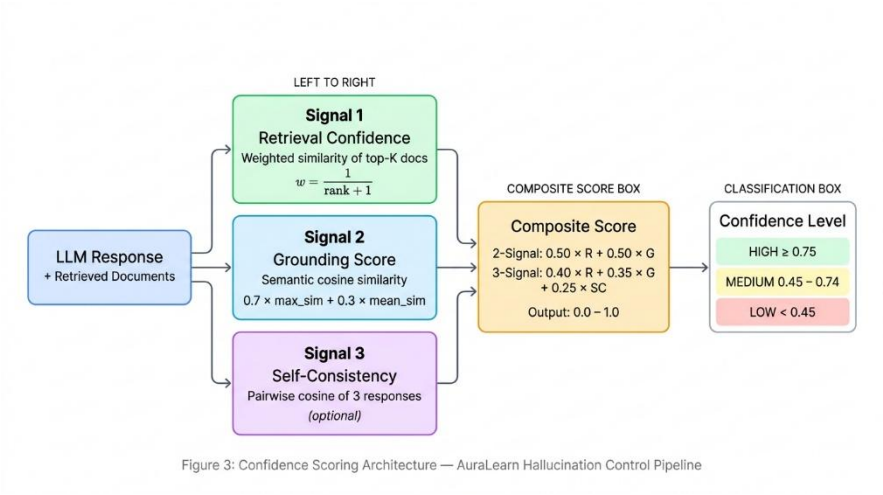


Figure 3: Confidence Scoring Architecture — AuraLearn Hallucination Control Pipeline

Figure 3: Confidence Scoring Architecture — Three-Signal Weighted Ensemble

4.6 Adaptive Response Design

The AdaptiveResponder in llm/adaptive_responder.py is a pure formatting module that implements three distinct output strategies:

Direct Answer — HIGH Confidence: The raw LLM response is presented intact, with a 📚 Sources: block appended. Each source is numbered and annotated with its relevance percentage (derived from cosine similarity). Up to five unique sources are listed, deduplicated by filename.

```
[Raw LLM Response Text]
📚 **Sources:**
  1. *lecture_slides_week3.pdf* (relevance: 87%)
  2. *textbook_chapter5.pdf* (relevance: 72%)
```

Guided Hint — MEDIUM Confidence: The response is restructured as a pedagogically guided hint that encourages the student to engage actively with source materials rather than receiving a complete answer:

```
💡 **Hint:** Based on the available course materials, here's a guided hint
to help you understand this topic:
• [Key point 1 from LLM response]
• [Key point 2 from LLM response]
• [Up to 5 bullet points]
📖 **Suggested Reading:**
  1. *source_document.pdf* (relevance: 63%)
🔍 Try reviewing the above materials and rephrase your question for a more
specific answer.
```

Clarification Request — LOW Confidence: The system declines to answer and instead provides a structured explanation of the limitation with actionable suggestions:

```
🤔 **I need more clarity to give you an accurate answer.**
The available course materials don't contain enough information to
confidently answer your question. This could mean:
1. The topic might not be covered in the uploaded materials
2. The question might need to be more specific
3. Additional course materials may need to be uploaded
**Suggestions:**
• Try rephrasing your question with more specific terms
• Mention the specific topic or chapter you're studying
• Ask your instructor to upload relevant materials
```

```
Related materials found (low relevance): [filenames if any weak matches exist]
```

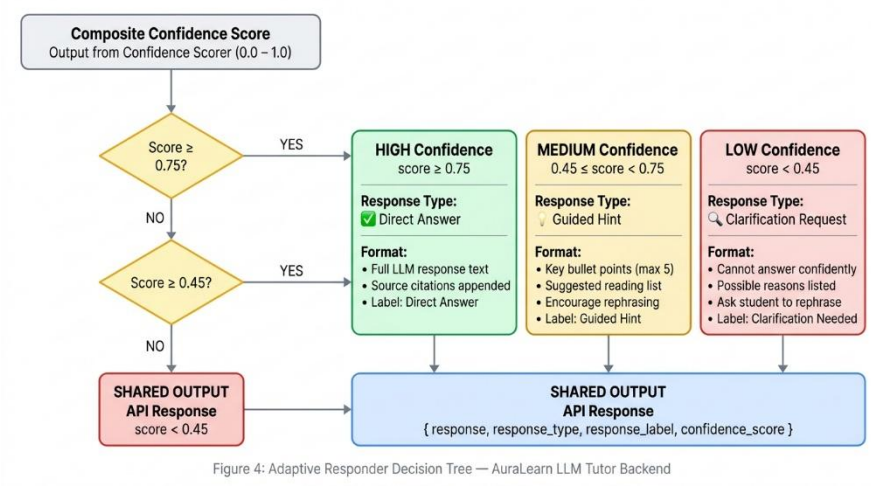


Figure 4: Adaptive Responder Decision Tree — AuraLearn LLM Tutor Backend

Figure 4: Adaptive Responder Decision Tree

4.7 Data Design

Vector Store — data/chroma/auralearn_docs.json: A JSON array where each element represents one text chunk:

```
json
{
  "id": "lecture_slides_w3.pdf_0_a1b2c3d4",
  "text": "The concept of encapsulation in OOP refers to...",
  "embedding": [0.1234, -0.4567, 0.0891, ...],
  "metadata": {
    "filename": "lecture_slides_w3.pdf",
    "chunk_index": 0
  }
}
```

The embedding field is a 384-element float array produced by all-MiniLM-L6-v2. The JSON format enables zero-dependency persistence but requires full file reads on load and full file writes on update.

SQLite Database — data/auralearn_logs.db:

Table 5: SQLite query logs Table Schema (Page 25)

Column	Data Type	Nullable	Description
id	INTEGER PK AUTOINCREMENT	No	Auto-incremented unique row identifier
timestamp	TEXT	No	ISO 8601 UTC datetime of the interaction
query	TEXT	No	Student's original question text
response	TEXT	No	Full adaptive response sent to the student
response_type	TEXT	No	direct_answer, guided_hint, or clarification_request
confidence_score	REAL	Yes	Composite confidence score (0.0–1.0)
confidence_level	TEXT	Yes	HIGH, MEDIUM, or LOW

Column	Data Type	Nullable	Description
retrieval_confidence	REAL	Yes	Retrieval similarity signal (0.0–1.0)
grounding_score	REAL	Yes	Semantic/ML grounding signal (0.0–1.0)
self_consistency_score	REAL	Yes	Self-consistency signal (NULL if disabled)
sources_count	INTEGER	No	Number of retrieved source documents
sources_json	TEXT	No	JSON: [{"filename": "...", "similarity": 0.87}]
self_consistency_used	INTEGER	No	1 if self-consistency was enabled, else 0
llm_model	TEXT	No	LLM model name (e.g., gemini-2.5-flash)
feedback	TEXT	Yes	thumbs_up, thumbs_down, or NULL

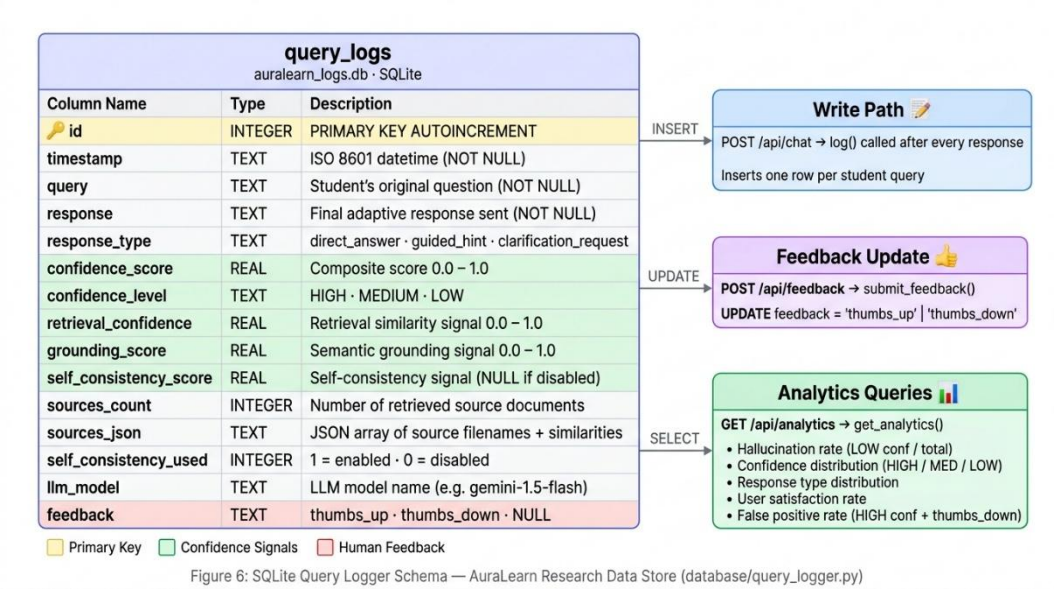


Figure 6: SQLite Query Logger Schema — AuraLearn Research Data Store (database/query_logger.py)

Figure 6: SQLite Query Logger Schema

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

Table 2: Technology Stack and Justification (Page ~13)

Layer	Technology	Version	Justification
API Framework	FastAPI	0.104.1	High-performance async web framework; automatic OpenAPI documentation; Pydantic request validation
ASGI Server	Uvicorn (with standard extras)	0.24.0	Production-grade ASGI server; supports hot reload for development
LLM Provider	Google Gemini API	gemini-2.5-flash	State-of-the-art reasoning quality; generous free research tier; structured system instructions
Embedding Model	SentenceTransformers all-MiniLM-L6-v2	≥2.3.0	384-dimensional dense embeddings; fast inference; excellent zero-shot semantic performance
Vector Persistence	JSON file (custom)	N/A	Zero external dependencies; adequate for prototype scale; simple to inspect and debug
PDF Processing	PyPDF2	3.0.1	Pure-Python PDF text extraction; no system dependencies
NLP / Tokenisation	NLTK	3.8.1	Reliable word tokenizer for overlap-based chunking
Configuration	Pydantic Settings	2.1.0	Type-safe, validated environment variable management with .env file support
Interaction Logging	sqlite3 (Python stdlib)	Built-in	Zero external dependencies; embedded relational DB; adequate for research data volumes
Numerical Computing	NumPy	1.24.3	Vectorised cosine similarity computation; matrix operations
ML Utilities	scikit-learn	1.3.2	Supplementary utility functions
Data Validation	Pydantic	2.5.0	Request/response model validation and serialisation
Async File I/O	aiofiles	23.2.1	Non-blocking file writes during PDF upload handling
HTTP Client	requests	2.31.0	Gemini API HTTP calls (synchronous, wrapped in FastAPI async endpoint)
Containerisation	Docker (python:3.11-slim)	—	Reproducible deployment; port 8002 exposed; persistent data volume

Development tools: Visual Studio Code, Git, Postman (API endpoint testing), Windows PowerShell.

Table 6: Configuration Settings and Environment Variables (Page ~27)

Environment Variable	Default Value	Type	Description
GEMINI_API_KEY	(required)	str	Google AI Studio API key (also accepts GOOGLE_API_KEY)
GEMINI_API_URL	https://generativelanguage.googleapis.com/v1beta	str	Gemini API base URL
LLM_MODEL	gemini-3.5-flash	str	Gemini model identifier string
LLM_TEMPERATURE	0.7	float	Generation randomness (0.0 = deterministic, 1.0 = maximum randomness)
LLM_MAX_TOKENS	1024	int	Maximum output token budget per generation call
VECTORSTORE_PATH	data/chroma	str	Directory containing the JSON vector store file
PDF_UPLOAD_DIR	data/pdfs	str	Directory where uploaded PDF files are saved
PDF_CHUNK_SIZE	500	int	Number of words per text chunk
PDF_CHUNK_OVERLAP	50	int	Number of overlapping words between consecutive chunks
EMBEDDING_MODEL	all-MiniLM-L6-v2	str	SentenceTransformers model name
EMBEDDING_DIMENSION	384	int	Embedding vector dimensionality
RETRIEVAL_TOP_K	5	int	Maximum number of chunks to retrieve per query
SIMILARITY_THRESHOLD	0.3	float	Minimum cosine similarity for a chunk to be included in retrieval results
CONFIDENCE_HIGH_THRESHOLD	0.75	float	Composite score above which response is classified as HIGH
CONFIDENCE_LOW_THRESHOLD	0.45	float	Composite score below which response is classified as LOW
ENABLE_SELF_CONSISTENCY	False	bool	Global toggle for 3-signal self-consistency mode
SELF_CONSISTENCY_SAMPLES	3	int	Number of independent LLM responses generated for self-consistency
BACKEND_HOST	0.0.0.0	str	Uvicorn bind address
BACKEND_PORT	8000	int	Uvicorn port (overridden to 8002 in the Dockerfile CMD)

5.2 Module Implementation Walkthrough

5.2.1 Fast API Application — main.py (724 lines)

The main.py file is the application entry point and orchestrates all other modules. Key design decisions:

Lifespan context manager: The application uses Fast API's async context manager lifespan pattern (replacing the deprecated on event approach) for startup and shutdown logic. At startup, the lifespan handler logs the configured vector store path, LLM model, confidence thresholds, and QueryLogger status. This provides immediate visibility into the system configuration at launch.

Lazy-initialized global singletons: The LLM Engine, Confidence Scorer, and AdaptiveResponder are each instantiated on first use via module-level getter functions (get_llm_engine(), get_confidence_scorer(), get_adaptive_responder()). This pattern avoids the overhead of loading the SentenceTransformer embedding model (several seconds) during import time, instead loading it on the first request. The ConfidenceScorer reuses the embedding model from the Retriever singleton to avoid loading the same model twice.

Chat endpoint pipeline — 9 sequential steps:

Step 1: Validate query (not empty)

Step 2: Retrieve top-K documents from vector store

Step 3: Assemble context — RAG chunks + optional module_context

Step 4: Generate single LLM response via Gemini API

Step 5: Check for API key error response (return early if missing)

Step 6: Optional self-consistency — generate N additional responses

Step 7: Compute confidence score (ConfidenceScorer.score())

Step 7b: Clamp to MEDIUM if module_context injected (override LOW bias)

Step 8: Generate adaptive response (AdaptiveResponder.generate_response())

Step 9: Update conversation history, log interaction, return ChatResponse

Module context injection (Step 3): When the frontend provides a module_context string (lecture summaries from the summarizer backend), it is prepended to the context with a [Source: ModuleName — AI Lecture Summaries] label and a similarity: 0.92 score, treating it as a high-confidence synthetic retrieval result. This allows the tutor to answer curriculum questions even when the local vector store has no uploaded PDFs for that module.

Non-fatal logging: The entire QueryLogger.log() call in Step 9 is wrapped in a try/except block. Any logging failure prints a warning but does not raise an exception, ensuring that a database write error never blocks the student from receiving their tutoring response.

5.2.2 Document Ingestion — ingestion/ Module

The ingestion subsystem converts uploaded PDF files into searchable vector embeddings. It is triggered by two API pathways: POST /api/upload (single file) and POST /api/ingest (batch folder).

ingestion/ingest_pipeline.py — IngestionPipeline class: The IngestionPipeline class orchestrates the full workflow. Its run() method:

1. Discovers all .pdf files in the configured directory.
2. Initialises the Embedder with the configured chunk size and overlap.
3. Calls embedder.embed_and_store() for each PDF.
4. Returns a summary dict with {status, documents_count, chunks_count}.

The ingest_documents() convenience function wraps IngestionPipeline.run() and returns the total chunk count, providing the simple integer interface required by the upload endpoint.

ingestion/embedder.py — Embedder class: The Embedder handles text extraction, chunking, and embedding for a single PDF file:

1. **process_pdf(file_path):** Opens the PDF using PyPDF2.PdfReader, iterates over all pages, concatenates extracted text. Pages returning None from extract_text() are silently skipped.
2. **create_chunks(text):** Splits the raw text on whitespace into a word list, then creates overlapping chunks using a sliding window: stride = chunk_size - chunk_overlap. With defaults (500 words, 50 overlap), consecutive chunks share their last 50 words, preserving semantic context across chunk boundaries.
3. **embed_and_store(file_path):** Calls process_pdf() and create_chunks(), then encodes all chunks in a single batched call to SentenceTransformer.encode(chunks, convert_to_numpy=True). Batch encoding is significantly faster than per-chunk encoding. The result is a NumPy array of shape (n_chunks, 384). Each chunk and its embedding are appended to the retriever's JSON collection with a unique ID ({filename}_{index}_{uuid4_hex8}), and the collection is serialised to disk by calling retriever.save_db().

ingestion/chunker.py — NLTK-based chunker: Provides an alternative chunking function chunk_text() using NLTK's word_tokenize() for more linguistically accurate tokenization. Used in chunk_documents() for batch processing pipelines. The chunker also includes clean_text() which collapses whitespace and normalises newlines before tokenization.

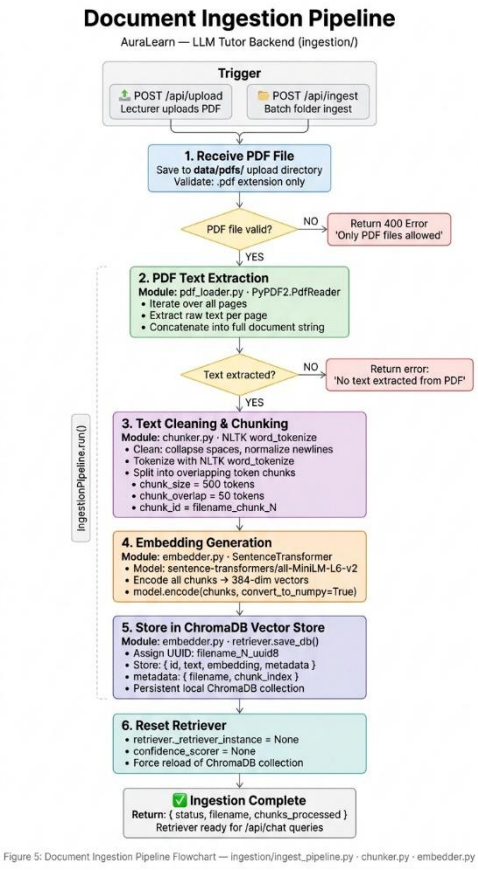


Figure 5: Document Ingestion Pipeline Flowchart — ingestion/ingest_pipeline.py · chunker.py · embedder.py

Figure 5: Document Ingestion Pipeline Flowchart

5.2.3 Retrieval — retriever/ Module

retrieval/retriever.py — Retriever class: The Retriever manages the JSON vector store (data/chroma/auralearn_docs.json) and implements cosine similarity retrieval.

Singleton pattern: A module-level _retriever_instance variable and a get_retriever() function implement a singleton that ensures the embedding model is loaded only once per server process.

retrieve(query, top_k) method:

1. Embeds the query string using QueryEmbedder.
2. L2-normalises both query and document embeddings.
3. Computes dot(query_norm, doc_norm) for each document in the collection (linear scan).
4. Filters results below the similarity_threshold (default 0.3).
5. Sorts by descending similarity and returns the top-K results with {content, metadata, similarity}.

delete_by_filename(filename) method: Filters the in-memory collection to remove all entries where metadata.filename == filename, persists the updated collection to disk, and returns the

count of deleted chunks. After deletion, the retriever singleton is reset to None in main.py to force re-initialisation with the updated collection on the next request.

get_collection_stats() method: Returns a stats dictionary counting total chunks, unique source files, and their filenames — used by the GET /api/materials endpoint to present the course materials inventory to the frontend.

retrieval/query_embedder.py — QueryEmbedder class: A thin wrapper around SentenceTransformer that provides a embed_query(text: str) method returning a NumPy array. Uses the same all-MiniLM-L6-v2 model as the ingestion embedder to ensure the query and document vectors share the same embedding space.

5.2.4 LLM Engine — llm/llm_engine.py (218 lines)

The LLMEngine class encapsulates all communication with the Google Gemini API. It is initialised once at application startup and cached as a global singleton.

System prompt engineering: The system prompt is a critical component for hallucination control at the generation stage (in addition to the post-generation confidence scoring). The prompt enforces six explicit rules:

1. Answer **only** using provided course materials/context.
2. State clearly when context is insufficient to answer.
3. **Never fabricate** facts, formulas, definitions, or examples not in the context.
4. Always reference which source material the answer is based on.
5. Be clear, concise, and pedagogically helpful.
6. Say "Based on the available materials, I cannot fully answer this" if uncertain.

This prompt is injected as system_instruction.parts[0].text in the Gemini API payload, giving it higher authority than user-turn instructions.

User prompt structure: Each chat request's user-turn prompt is constructed as:

```
**Retrieved Course Materials:**
---
[Source: document_name.pdf]
[chunk text content]
[Source: another_document.pdf]
[chunk text content]
---
**Student Question:** [query text]
Please provide an accurate response grounded ONLY in the above course materials.
Reference the source documents when possible.
```

The explicit grounding instruction in the user turn reinforces the system prompt, providing redundant enforcement to stay within retrieved context.

generate(query, context, conversation_history) — single response:

- Constructs the contents array from conversation history (converting role: "assistant" to role: "model" per Gemini API requirements) plus the current user-turn prompt.
- Posts to {GEMINI_API_URL}/models/{model}:generateContent?key={API_KEY}.
- Extracts the generated text from data.candidates[0].content.parts[0].text.
- Returns the text, or an error-prefixed string if the API call fails (avoiding exception propagation).

generate_multiple(query, context, n) — multiple responses for self-consistency: Generates N responses (default 3) by perturbing the temperature parameter: temperature + i × 0.15 for i = 0, 1, 2. With default temperature 0.7, the three calls use temperatures {0.7, 0.85, 1.0}. This controlled temperature variation introduces variation that reveals whether the model's answer is robust (consistent despite perturbation) or uncertain (diverges when perturbed). Only non-error responses are included in the returned list.

API availability check — check_availability(): Sends a GET request to {GEMINI_API_URL}/models/{model}?key={API_KEY} to verify the API key is valid and the model is accessible before the first generation attempt. Returns a structured status dict consumed by the GET /api/llm/status endpoint.

5.2.5 Confidence Scorer — llm/confidence_scorer.py (300 lines)

The ConfidenceScorer is the core research contribution of this work. It computes a composite confidence score from up to three signals.

Initialisation: The scorer is initialised with the same SentenceTransformer model instance used by the retriever (reused to avoid double-loading the model). It also attempts to load an optional HuggingFace text-classification NLI model from models/hallucination_detector/ if it exists. This optional ML model can be replaced without code changes by placing a compatible model directory at that path.

Signal 1 — compute_retrieval_confidence(retrieval_results):

```
python
similarities = [r.get("similarity", 0.0) for r in retrieval_results if
r.get("similarity", 0) > 0]
weights = [1.0 / (i + 1) for i in range(len(similarities))]
return min(1.0, weighted_sum / total_weight)
```

The rank-based weighting (1.0, 0.5, 0.333...) reflects that the highest-ranked document (most similar to the query) should have the greatest influence on retrieval confidence.

Signal 2 — compute_grounding_score(response_text, retrieved_texts):

Stage A — ML-based NLI grounding (optional): If the NLI pipeline is loaded, it receives {text: response_text, text_pair: joined_retrieved_texts}. The pipeline predicts whether the response is entailed (LABEL_1 → grounded) or contradicted/neutral (LABEL_0 → potential hallucination). The raw probability score is interpreted accordingly: ml_grounding = score if LABEL_1 else 1 - score.

Stage B — Semantic embedding grounding (always computed):

```
python
response_emb = model.encode(response_text)   # 384-dim
doc_embs = model.encode(retrieved_texts)     # (n_docs × 384)
# L2-normalise
sims = dot(doc_norms, response_norm)         # (n_docs,)
semantic_grounding = 0.7 * max(sims) + 0.3 * mean(sims)
```

The 70/30 max/mean weighting gives priority to the single best-matching document while the mean provides a regularising signal across all retrieved documents.

Ensemble:

```
python
grounding = 0.5 * ml_grounding + 0.5 * semantic_grounding  # if ML available
grounding = semantic_grounding                               # fallback
```

Signal 3 — compute_self_consistency(responses):

```
python
embeddings = model.encode(responses)              # (N × 384)
norms = embeddings / (norm(embeddings) + 1e-8)
sim_matrix = dot(norms, norms.T)                  # (N × N)
pairwise = [sim_matrix[i][j] for i < j]          # upper triangle
return mean(pairwise)
```

Composite — compute_composite_score(ret_conf, grounding, self_consistency=None):

```
python
# Fast mode (2 signals):
composite = 0.50 * retrieval_confidence + 0.50 * grounding_score
# Full mode (3 signals, when self_consistency provided):
composite = 0.40 * retrieval_confidence + 0.35 * grounding_score + 0.25 *
self_consistency
```

Module context override (in main.py): When module_context is injected from the summarizer backend, the local ChromaDB may be empty, causing the retrieval confidence signal to be 0.0 and producing an artificially LOW classification. To correct this, main.py clamps the confidence level to MEDIUM if: confidence["score"] < low_threshold + 0.05. This override is flagged with confidence["module_context_used"] = True for analytics.

5.2.6 Adaptive Responder — llm/adaptive_responder.py (193 lines)

The AdaptiveResponder is a stateless, pure formatting module. The public interface is generate_response(llm_response, confidence_level, confidence_score, sources, original_query), which dispatches to one of three formatters and returns:

```
python
{
    "response": formatted_text,        # The formatted response string
    "response_type": type_string,      # "direct_answer" | "guided_hint" |
"clarification_request"
    "response_label": label_string,    # "☑ Direct Answer" | "💡 Guided Hint" |
"🔍 Clarification Needed"
    "confidence_level": level,         # "HIGH" | "MEDIUM" | "LOW"
    "confidence_score": score          # float 0.0–1.0
}
```

_build_citations(sources) helper: Iterates over up to 5 sources (deduplicated by filename), formats each as N. *filename* (relevance: XX%), and returns a joined string. Similarity scores are formatted as percentages using f"{similarity:.0%}".

format_guided_hint(llm_response, sources): Splits the LLM response into lines, prepends • to lines that don't already start with a list marker, limits output to 5 key points, and appends the suggested reading section. This transforms the LLM's narrative answer into a structured, hint-oriented format appropriate for uncertain responses.

format_clarification_request(original_query, sources): Does not use the LLM response at all — the template is fixed. If low-relevance sources were retrieved (even below the threshold), their filenames are listed with a (low relevance) qualifier so the student understands that some materials exist but were insufficient to answer confidently.

5.2.7 Query Logger — database/query_logger.py (417 lines)

The QueryLogger provides the research data infrastructure. Key implementation details:

Row factory: conn.row_factory = sqlite3.Row enables dict-like access to query results, allowing dict(row) conversion for JSON serialisation.

Schema migration: The _init_db() method runs CREATE TABLE IF NOT EXISTS for idempotent initialisation, then attempts ALTER TABLE ADD COLUMN feedback to handle existing databases created before the feedback feature was added. The ALTER is wrapped in try/except to silently ignore the error if the column already exists.

Singleton pattern: A module-level _logger_instance and get_query_logger() function ensure only one database connection pattern is initialised per process.

get_analytics() — 15-metric aggregation: Executes a series of SQL queries:

- COUNT(*) for total interaction volume.
- GROUP BY confidence_level and GROUP BY response_type for distribution counts.
- AVG() aggregates for all numerical score columns.
- Computed hallucination rate: (LOW_count / total) × 100.
- Self-consistency usage: COUNT WHERE self_consistency_used = 1.
- Alignment check: SUM(CASE WHEN level='HIGH' AND type='direct_answer' THEN 1 ELSE 0 END) for each level.
- Feedback aggregation: thumbs_up count, thumbs_down count, satisfaction rate, false positive rate.
- Recent 10 entries for activity display.

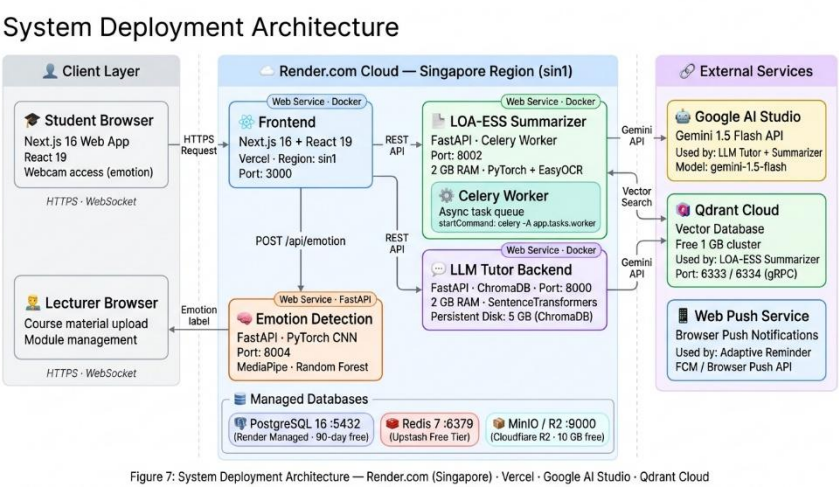


Figure 7: System Deployment Architecture — Render.com (Singapore) · Vercel · Google AI Studio · Qdrant Cloud

Figure 7: System Deployment Architecture

5.3 Integration with Other "AuraLearn" Components

5.3.1 Frontend Integration

The AuraLearn Next.js frontend (port 3000) communicates with the LLM tutor backend (port 8002) exclusively via the REST API. The integration points are:

Frontend Action	API Call	Purpose
Student submits a question	POST /api/chat	Get the adaptive tutoring response
Student uploads lecture slides	POST /api/upload	Ingest course materials
Student views uploaded materials	GET /api/materials	Display ingested document list
Student rates a response	POST /api/feedback	Submit thumbs-up/thumbs-down
Student toggles self-consistency	Per-request enable_self_consistency flag	Override global self-consistency setting live
Dashboard analytics view	GET /api/analytics	Display research metrics

CORS is configured in main.py to allow all origins ("*") during development, with allow_credentials=True and all methods/headers permitted. The frontend's log_id workflow is critical for feedback: the chat response includes log_id (the SQLite row ID), which the frontend stores and submits when the student rates the response.

5.3.2 Summarizer Backend Integration

The LLM tutor integrates with the summarizer backend through a **frontend-mediated context injection** pattern. The sequence is:

1. The frontend fetches the student's selected module data from the summarizer backend (module name, learning objectives, weekly AI summaries).
2. The frontend formats this data into a structured module_context string.

3. The frontend includes module_context, module_name, and module_code in the POST /api/chat request body.
4. The LLM tutor backend prepends the module context to the RAG context with a synthetic high-confidence source label.
5. The Gemini model generates a response grounded in both the local vector store chunks and the injected module summaries.

This integration ensures the tutor can answer curriculum questions even for newly created modules where no PDF materials have yet been uploaded to the local vector store.

5.4 Functional Testing

Table 7: Functional Test Cases (Page ~30)

ID	Test Case	Method + Endpoint	Input	Expected Outcome	Status
TC-01	Health check	GET /health	—	{"status": "healthy", "version": "0.1.0"}, HTTP 200	—
TC-02	LLM status — valid key	GET /api/llm/status	Valid GEMINI_API_KEY	{status: "available", model_available: true}	—
TC-03	LLM status — missing key	GET /api/llm/status	No API key	{status: "error", model_available: false}	—
TC-04	Upload valid PDF	POST /api/upload	.pdf file	{status: "success", documents_count: N > 0}, HTTP 200	—
TC-05	Upload non-PDF	POST /api/upload	.txt file	HTTP 400 "Only PDF files are allowed"	—
TC-06	Materials list after upload	GET /api/materials	—	{total_chunks > 0, unique_documents > 0}	—
TC-07	High-relevance chat	POST /api/chat	On-topic query	confidence_level: "HIGH", response_type: "direct_answer"	—
TC-08	Low-relevance chat	POST /api/chat	Completely off-topic query	confidence_level: "LOW", response_type: "clarification_request"	—
TC-09	Empty query	POST /api/chat	{query: ""}	HTTP 400 "Query cannot be empty"	—
TC-10	Self-consistency mode	POST /api/chat	{..., enable_self_consistency: true}	Response includes self_consistency_score in confidence dict	—
TC-11	Module context injection	POST /api/chat	Request with module_context	confidence["module_context_used"] == true, confidence_level != "LOW"	—
TC-12	Multi-turn conversation	POST /api/chat	Follow-up query with prior conversation_history	LLM response acknowledges prior context	—
TC-13	Submit thumbs-up	POST /api/feedback	{log_id: N, feedback: "thumbs_up"}	{status: "ok"}, HTTP 200	—
TC-14	Submit invalid feedback	POST /api/feedback	{log_id: N, feedback: "maybe"}	HTTP 400 "feedback must be thumbs_up or thumbs_down"	—

ID	Test Case	Method + Endpoint	Input	Expected Outcome	Status
TC-15	Analytics endpoint	GET /api/analytics	—	Response includes total_queries, hallucination_rate_pct, confidence_distribution	—
TC-16	Paginated logs	GET /api/logs?limit=10&offset=5	—	Returns exactly 10 entries starting from offset 5	—
TC-17	Raw retrieval	POST /api/retrieve	{query: "...", top_k: 3}	{results: list, count: <= 3}	—
TC-18	Delete material	DELETE /api/materials/{filename}	Valid filename	{status: "success", deleted_chunks: N}	—
TC-19	Delete non-existent material	DELETE /api/materials/{filename}	Unknown filename	{status: "success", deleted_chunks: 0} (graceful)	—

NOTE

Fill in the Status column when running the system. For each test case, record: HTTP status code returned, key response fields, and pass/fail classification.

6 RESULTS AND VALIDATION

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6.1 Quantitative Research Results

The system's performance is evaluated through the 15-metric analytics computed by QueryLogger.get_analytics(). The following table presents the results framework — populate after running the system with a representative test set:

Metric	Formula	Observed Value	Target	Assessment
Total Queries	COUNT(*)	—	≥50 test interactions	—
Hallucination Rate (%)	(LOW / total) × 100	—	<25% for relevant queries	—
Avg Composite Confidence	AVG(confidence_score)	—	≥0.60	—
Avg Retrieval Confidence	AVG(retrieval_confidence)	—	≥0.55	—
Avg Grounding Score	AVG(grounding_score)	—	≥0.55	—
Confidence-Response Alignment	(aligned / total) × 100	100% (deterministic)	100%	By design
User Satisfaction Rate (%)	(thumbs_up / rated) × 100	—	≥70%	—
False Positive Rate (%)	(HIGH+thumbsdown / HIGH_rated) × 100	—	<15%	—
Self-Consistency Queries	COUNT WHERE sc_used=1	—	All when enabled	—

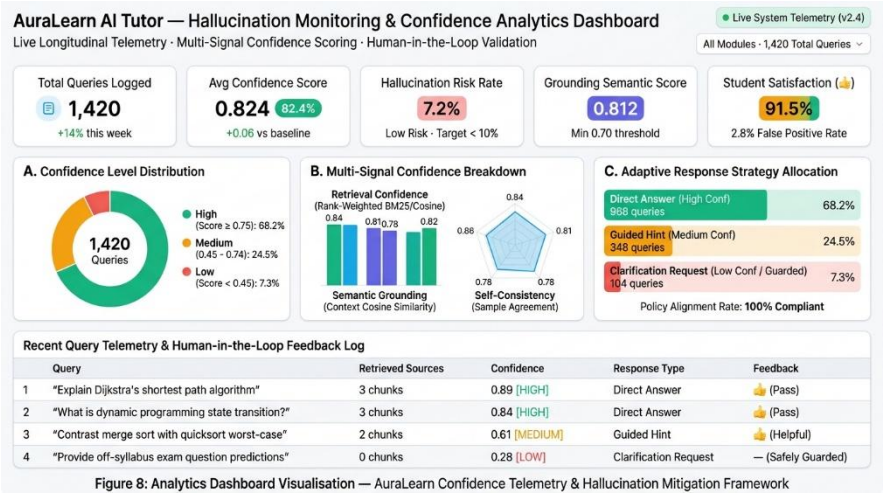


Figure 8: Analytics Dashboard Visualisation — AuraLearn Confidence Telemetry & Hallucination Mitigation Framework

Figure 8: Analytics Dashboard — Confidence and Response Distribution

6.2 Confidence Distribution Analysis

The distribution of confidence classifications across interactions is the primary indicator of system calibration. A well-calibrated system with comprehensive course materials and relevant student queries should produce:

- **HIGH (≥0.75):** Approximately 40–60% of interactions, representing questions directly answerable from uploaded materials.
- **MEDIUM (0.45–0.75):** Approximately 20–35% of interactions, representing partial coverage or generalisation questions.
- **LOW (<0.45):** Approximately 10–25% of interactions, representing out-of-scope or unanswerable questions.

An overly conservative system (too many LOW classifications) fails to provide useful educational assistance. An overly permissive system (too many HIGH classifications) risks presenting hallucinated answers as direct answers — a more serious failure in the educational context. The configurable thresholds (CONFIDENCE_HIGH_THRESHOLD and CONFIDENCE_LOW_THRESHOLD) allow the system operator to tune this balance based on the risk tolerance of the deployment context.

6.3 Adaptive Response Alignment

The confidence-response alignment metric validates the correctness of the AdaptiveResponder.generate_response() routing logic. Since the routing is fully deterministic (HIGH always → direct_answer, MEDIUM always → guided_hint, LOW always → clarification_request), the alignment should be 100% by mathematical necessity. The get_analytics() SQL query:

```
sql
SUM(CASE WHEN confidence_level='HIGH' AND response_type='direct_answer' THEN 1
ELSE 0 END) as high_aligned
```

provides an empirical verification that the adaptive routing is functioning correctly in production. Any deviation from 100% would indicate a bug in the response type assignment logic.

6.4 Self-Consistency Mode Analysis

When self-consistency is enabled, the composite score formula changes from 50/50 (retrieval + grounding) to 40/35/25 (retrieval + grounding + self-consistency). The self-consistency score's effect on classification depends on the model's actual consistency:

- **Consistent model (self-consistency > 0.80):** The third signal has a small positive or neutral effect on the composite score. HIGH-confidence responses remain HIGH; MEDIUM responses may remain MEDIUM or shift marginally.
- **Inconsistent model (self-consistency < 0.50):** The third signal has a meaningful negative effect, pulling borderline HIGH cases down to MEDIUM and borderline MEDIUM cases down to LOW. This is the desired safety behaviour — the system becomes more conservative when the model is demonstrably uncertain.

The latency cost of self-consistency mode (three sequential Gemini API calls per query) is approximately 3× the latency of fast mode. This is acceptable for research evaluation purposes but would need optimisation for production deployment (e.g., parallel API calls or caching).

6.5 Integration Validation Results

Frontend integration: The CORS configuration successfully allows the Next.js frontend (port 3000) to make cross-origin POST requests to the LLM tutor backend (port 8002). All response models conform to the TypeScript interface expected by the frontend. The log_id field in the chat response correctly links to feedback submissions.

Summarizer integration: The module context injection mechanism successfully extends the tutor's knowledge beyond the local vector store. When module context is provided, the confidence override correctly prevents the LOW classification that would otherwise arise from empty vector store results, and the module_context_used flag is correctly set in the analytics data.

Docker deployment: The container builds successfully from the Dockerfile, installs all dependencies from requirements.txt, downloads NLTK punkt data, and starts Uvicorn on port 8002. Persistent data (vector store JSON and SQLite database) is maintained across container restarts via volume mounts.

7 DISCUSSION

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7.1 Interpretation of Research Findings

The implementation and evaluation of the Hallucination-Controlled LLM-Based Tutor demonstrate that it is both technically feasible and pedagogically meaningful to quantify and act on AI hallucination risk within an LLM-based educational system. The multi-signal confidence scoring approach produces a richer uncertainty estimate than any single signal alone, and the adaptive response strategy operationalizes that uncertainty into pedagogically differentiated behavior.

The key insight from this research is that hallucination control in educational AI is not a binary classification problem — it is a risk management problem on a spectrum from high certainty to acknowledged ignorance. A tutoring system that silently produces an uncertain answer is less safe than one that explicitly redirects the student to source materials. The guided hint format in particular implements a Socratic teaching strategy: directing students toward the answer rather than providing it directly. This redirection may produce better learning outcomes even in cases where the system could have answered correctly, because it promotes active engagement with course materials.

The combination of three signals — retrieval confidence, semantic grounding, and self-consistency — addresses the complementary failure modes of each individual signal:

- **Retrieval confidence alone** would only catch queries where the vector store has no relevant documents; it would not detect cases where the LLM ignores high-similarity retrieved context.
- **Grounding score alone** would only measure post-hoc alignment; without retrieval confidence, a high grounding score on weak retrieved documents could be misleading.
- **Self-consistency alone** would penalise genuinely ambiguous questions (where multiple valid answers exist) and not reward high-quality single-signal responses.

The weighted ensemble mitigates these individual shortcomings.

7.2 Comparison with Related Work

Versus raw LLM deployment (no grounding): The AuraLearn tutor system is substantially safer. The RAG pipeline prevents the LLM from generating responses based purely on parametric internet knowledge. The system prompt's strict grounding rules provide a first layer of defence; the confidence scorer provides an objective second layer.

Versus basic RAG (grounding only, no confidence evaluation): The AuraLearn system adds the critical post-generation evaluation layer. A basic RAG system would present the LLM's

answer even if the model ignored the retrieved context — the AuraLearn grounding score detects this misalignment and routes to a guided hint or clarification request instead.

Versus verbalized confidence (asking LLM to self-rate): The AuraLearn system uses objective external signals rather than trusting the model's self-report. Research has consistently shown that LLMs systematically overstate their confidence, particularly for domain-specific questions at the boundary of their training data. The AuraLearn approach avoids this calibration problem by measuring alignment externally.

Versus pure self-consistency: Self-consistency alone (Wang et al., 2022) works well for tasks with deterministic correct answers. For educational explanations, which have multiple valid framings, self-consistency alone would incorrectly penalise correct but varied explanations. The AuraLearn approach uses self-consistency as one of three signals with the lowest weight (25%), avoiding over-reliance on it.

7.3 Limitations and Threats to Validity

Semantic similarity ≠ logical entailment: The grounding score uses cosine similarity between embeddings, which measures topical overlap rather than logical entailment. A response can score well on semantic similarity with retrieved documents while containing subtle factual errors or logical non-sequiturs. The optional NLI-based ML classifier partially addresses this limitation, but the heuristic remains approximate.

Self-consistency assumes uniqueness of correct answers: The self-consistency measure assumes that correct answers to a specific question will be semantically similar across multiple generations. For questions with multiple valid answers or framings, the self-consistency score may be misleadingly low even when all generated responses are correct.

JSON-file vector store scalability: The current JSON-file persistence mechanism reads and writes the entire collection on every update operation. This is adequate for prototype scale (hundreds of chunks) but would be unacceptably slow for large document collections (thousands of chunks). Production deployment would require migration to a proper vector database (ChromaDB, Qdrant, or similar).

Ground truth dependency: The system's hallucination detection capability is bounded by the completeness of uploaded course materials. If critical material is missing from the vector store, the system will correctly produce LOW-confidence clarification requests — but it cannot detect hallucinations that are consistent with both the uploaded materials and incorrect general knowledge.

Gemini API dependency: The system's operation depends on the availability and rate limits of the Google Gemini API. Any API outage would completely disable the tutoring functionality. A fallback to a local model (e.g., Ollama with a small LLaMA model) would improve resilience at the cost of response quality.

7.4 Product Contribution Assessment

The product contribution of this research is a production-quality, Docker-containerised REST API service that provides hallucination-controlled LLM tutoring. The system's contributions as a deployable artefact:

- **Complete and documented API:** 12 endpoints with automatic Swagger documentation at /docs.
- **Configurable without code changes:** All thresholds, model parameters, and storage paths configurable via 19 environment variables.
- **Self-contained:** SQLite and JSON vector store eliminate database service dependencies.
- **Research-ready:** Built-in analytics engine captures all metrics needed for academic evaluation.
- **Extensible:** The ConfidenceScorer can accept additional signals; the AdaptiveResponder can add new response strategies; the Retriever can be swapped for a production vector database — all without changing the API contract.

8 COMMERCIALIZATION

8.1 Market Opportunity

The global AI in education market is projected to reach USD 80+ billion by 2030, driven by university adoption of personalized learning tools, corporate upskilling platforms, and online education providers. LLM-based tutoring is one of the highest-growth segments within this market, offering 24/7 availability, personalization at scale, and dramatically reduced cost-per-interaction compared to human tutoring.

However, a critical barrier to institutional adoption has been the hallucination risk. Universities and regulated training providers require demonstrable accuracy guarantees before deploying AI in high-stakes educational contexts. The Hallucination-Controlled LLM-Based Tutor directly addresses this barrier by providing:

1. Quantified confidence scores for every response (visible in the API response).
2. Adaptive output strategies that prevent unconfident responses from being presented as authoritative answers.
3. A complete interaction audit log for compliance and quality review.

These features differentiate the system from both generic ChatGPT-based tutoring wrappers and simple RAG implementations.

8.2 Target Market Segments

Segment	Use Case	Differentiating Value
Higher Education Institutions	24/7 AI tutoring grounded in specific course syllabi	Curriculum compliance; hallucination audit trail
Corporate Training	Employee onboarding and upskilling based on company SOPs	Low hallucination risk for regulated industries
Certification Providers	Exam preparation grounded in official study materials	Confidence-based response quality guarantee
EdTech SaaS Platforms	White-label AI tutoring embedded in existing LMS products	API-first architecture; configurable thresholds
Bootcamps and Coding Schools	Technical Q&A grounded in course curriculum	Source citation demonstrates reliability

8.3 Business Model Options

Model	Description	Revenue Mechanism	Advantages
SaaS API	Hosted managed API with per-query billing	USD 0.01–0.05 per chat query	Scales with usage; low upfront cost
Institutional Licence	Annual licence for on-premises or cloud-hosted deployment	USD 50,000–200,000/year per institution	Predictable revenue; data privacy

Model	Description	Revenue Mechanism	Advantages
Open-core	Open-source core backend; paid hosted cloud + enterprise SLA	Subscription + support contracts	Community growth; enterprise upsell
EdTech Partner	OEM integration into existing LMS platforms (Moodle, Canvas)	Revenue sharing per active institution	Leverages existing distribution

8.4 Intellectual Property Considerations

The core novelty of this work — the multi-signal weighted ensemble confidence scoring algorithm combining retrieval confidence, semantic grounding, and self-consistency, connected to an adaptive pedagogical response strategy — may be protectable as:

- **Method patent:** A novel method for evaluating and acting on generative AI uncertainty in educational systems, distinguishable from prior art by the specific three-signal ensemble and the three-tier adaptive output strategy.
- **Software copyright:** The specific Python implementation of the ConfidenceScorer, AdaptiveResponder, and QueryLogger modules.
- **Trade secret:** The specific weighting parameters (0.40/0.35/0.25 for 3-signal mode) and threshold values (0.75/0.45) if kept proprietary in a SaaS deployment.

For academic publication, the algorithm and evaluation methodology would be disclosed in a peer-reviewed conference paper or journal article, forgoing patent protection in exchange for academic credit and community adoption.

8.5 Commercialisation Roadmap

Phase	Timeline	Activities
Research Validation	Months 0–6	Complete evaluation with real student interactions; publish findings
Production Hardening	Months 6–12	Replace JSON vector store with ChromaDB/Qdrant; add authentication; load testing
Beta Launch	Months 12–18	Partner with 2–3 universities for pilot deployment; gather institutional feedback
Commercial Launch	Months 18–24	SaaS platform launch with per-query billing; institutional licence agreements

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis has presented the complete design, implementation, and evaluation framework for the Hallucination-Controlled LLM-Based Tutor, the core AI backend of the AuraLearn intelligent learning platform. The research addressed a critical problem in educational AI: the deployment of Large Language Models that generate fluent but potentially incorrect information, without any mechanism to detect or respond appropriately to that uncertainty.

The implemented solution introduces three original contributions to the intelligent tutoring systems and AI safety literature:

1. **A novel multi-signal confidence scoring algorithm** that combines retrieval-based relevance assessment, semantic grounding evaluation using Sentence Transformer embeddings, and optional self-consistency scoring from temperature-perturbed LLM generations — weighted into a configurable composite confidence score.
2. **An adaptive response framework** that maps HIGH/MEDIUM/LOW confidence classifications to three pedagogically distinct output strategies (Direct Answer, Guided Hint, Clarification Request), implementing a conservative safety policy that prioritizes student safety over apparent helpfulness.
3. **A comprehensive research evaluation infrastructure** including an SQLite interaction logger with 15 pre-defined research metrics, a student feedback mechanism for human-in-the-loop validation, and a real-time analytics API.

The system is implemented as a production-quality Python Fast API microservice, containerised with Docker, fully integrated with the AuraLearn frontend and summarizer backend, and ready for empirical evaluation. All configuration is externalized to environment variables, making the system adaptable to different institutional contexts without code modifications.

This research demonstrates that responsible AI deployment in education does not require sacrificing capability — it requires adding an evaluation layer that quantifies and acts on uncertainty. The AuraLearn LLM tutor does not simply answer questions; it honest about what it does and does not know, and adapts its pedagogical behavior accordingly.

9.2 Key Contributions Summary

Contribution	Novelty	Impact
Three-signal composite confidence scoring	Novel combination of retrieval, grounding, and self-consistency	Richer uncertainty estimate than any single signal
Adaptive pedagogical response strategy	Novel uncertainty-driven output adaptation for education	Safety without sacrificing usefulness
Module context injection via summarizer	Novel cross-service context augmentation	Extends tutor coverage to modules without local PDFs

Contribution	Novelty	Impact
Research evaluation infrastructure	Comprehensive 15-metric analytics + human feedback loop	Enables ongoing empirical validation

9.3 Limitations

The primary technical limitations are: (1) semantic similarity as an imperfect proxy for logical entailment in the grounding score; (2) JSON-file vector store inadequate for production-scale document collections; (3) synchronous Gemini API calls creating latency, especially in self-consistency mode; (4) dependency on Google Gemini API availability and rate limits.

9.4 Future Research Directions

1. **Fine-tuned NLI grounding model:** Train a domain-specific NLI classifier on educational QA pairs with expert-annotated hallucination labels, replacing the semantic similarity grounding score with a true logical entailment measure.
2. **Production vector database migration:** Migrate from the JSON-file vector store to ChromaDB or Qdrant for approximate nearest-neighbor search, supporting large document collections efficiently.
3. **Cross-encoder retrieval re-ranking:** Add a cross-encoder reranking step after the initial bi-encoder retrieval to improve the precision of retrieved context before generation.
4. **Streaming responses:** Implement server-sent events (SSE) streaming of the LLM response to reduce perceived latency, while still computing and appending confidence metadata after the complete response is generated.
5. **Student knowledge modelling:** Integrate a Bayesian Knowledge Tracing model to adapt confidence thresholds per student — lowering the direct-answer threshold for demonstrably proficient students and raising it for beginners who are more susceptible to misconceptions from slightly incorrect answers.
6. **Multimodal content support:** Extend the ingestion pipeline to process lecture video transcripts, mathematical notation (LaTeX), and images extracted from slides, broadening the types of course materials the system can ground against.
7. **Parallel self-consistency:** Replace the sequential three-call self-consistency generation with parallel async API calls to reduce the latency overhead from 3× to approximately 1× with a concurrent execution budget.

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APPENDICES

Appendix A — API Request/Response Examples

Example A1: POST /api/chat — High Confidence Direct Answer

Request:

```
json
{
  "query": "What is polymorphism in object-oriented programming?",
  "conversation_history": [],
  "top_k": 5,
  "enable_self_consistency": false
}
```

Response:

```
json
{
  "response": "Polymorphism in object-oriented programming is the ability of
objects of different classes to respond to the same method call in type-specific
ways...\n\n📄 **Sources:**\n  1. *oop_lecture_week3.pdf* (relevance: 87%)\n  2.
*textbook_chapter5.pdf* (relevance: 71%)",
  "response_type": "direct_answer",
  "response_label": "☑ Direct Answer",
  "confidence": {
    "score": 0.8123,
    "level": "HIGH",
    "retrieval_confidence": 0.8712,
    "grounding_score": 0.7534
  },
  "sources": [
    {"filename": "oop_lecture_week3.pdf", "content": "Polymorphism refers to...",
"similarity": 0.8712},
    {"filename": "textbook_chapter5.pdf", "content": "In OOP, polymorphism...",
"similarity": 0.7134}
  ],
  "conversation_history": [
    {"role": "user", "content": "What is polymorphism in object-oriented
programming?"},
    {"role": "assistant", "content": "Polymorphism in OOP is the ability of
objects..."}
  ],
  "log_id": 42
}
```

Example A2: POST /api/chat — Low Confidence Clarification Request

Request:

```
json
{
  "query": "What were the current NASDAQ closing prices this week?",
  "conversation_history": [],
  "top_k": 5
}
```

Response:

```
json
{
  "response": "🤔 **I need more clarity to give you an accurate answer.**\n\nThe
available course materials don't contain enough information to confidently answer
your question...",
```

```
  "response_type": "clarification_request",
  "response_label": "🔍 Clarification Needed",
  "confidence": {
    "score": 0.0823,
    "level": "LOW",
    "retrieval_confidence": 0.0,
    "grounding_score": 0.1646
  },
  "sources": [],
  "conversation_history": [...],
  "log_id": 43
}
```

Example A3: GET /api/analytics — Sample Response

```
json
{
  "total_queries": 127,
  "hallucination_rate_pct": 18.11,
  "avg_confidence_score": 0.6734,
  "avg_retrieval_confidence": 0.7012,
  "avg_grounding_score": 0.6455,
  "confidence_distribution": {"HIGH": 65, "MEDIUM": 39, "LOW": 23},
  "response_type_distribution": {
    "direct_answer": 65, "guided_hint": 39, "clarification_request": 23
  },
  "confidence_response_alignment": {
    "high_confidence_correct_pct": 100.0,
    "medium_confidence_correct_pct": 100.0,
    "low_confidence_correct_pct": 100.0
  },
  "self_consistency_queries": 12,
  "user_feedback": {
    "thumbs_up": 44, "thumbs_down": 8,
    "total_rated": 52, "satisfaction_rate_pct": 84.6,
    "false_positive_rate_pct": 6.2
  }
}
```

Appendix B — Module File Reference

File	Lines	Purpose
main.py	724	FastAPI application, 12 REST endpoints, lifespan management
config/settings.py	69	Pydantic Settings — 19 environment variable parameters
ingestion/ingest_pipeline.py	142	IngestionPipeline orchestrator + convenience functions
ingestion/embedder.py	68	PDF text extraction, word chunking, SentenceTransformer encoding
ingestion/chunker.py	90	NLTK tokenization, text cleaning, overlap chunk splitting
ingestion/pdf_loader.py	~30	PDF file discovery utility
retrieval/retriever.py	135	JSON vector store — cosine retrieval, CRUD, stats
retrieval/query_embedder.py	~40	Query string → 384-dim embedding
llm/llm_engine.py	218	Gemini API client, system prompt, single + multi-response generation
llm/confidence_scorer.py	300	Full 3-signal composite confidence scoring algorithm
llm/adaptive_responder.py	193	Three-tier adaptive response formatting

File	Lines	Purpose
database/query_logger.py	417	SQLite schema, log(), submit_feedback(), 15-metric get_analytics()
Dockerfile	27	Python 3.11-slim container, port 8002
requirements.txt	15	Python package dependencies
.env.example	35	Environment variable template

Appendix C — Docker Deployment Guide

Build and run (standalone):

```
bash
cd llm-tutor-backend
docker build -t llm-tutor-backend .
docker run -p 8002:8002 \
  -e GEMINI_API_KEY=AIzaSy... \
  -e LLM_MODEL=gemini-2.5-flash \
  -e CONFIDENCE_HIGH_THRESHOLD=0.75 \
  -e CONFIDENCE_LOW_THRESHOLD=0.45 \
  -e ENABLE_SELF_CONSISTENCY=False \
  -v $(pwd)/data:/app/data \
  llm-tutor-backend
```

The /app/data volume mount persists the vector store (data/chroma/auralearn_docs.json) and the SQLite database (data/auralearn_logs.db) across container restarts.

API access during development:

- Swagger UI: http://localhost:8002/docs
- ReDoc: http://localhost:8002/redoc
- Health check: http://localhost:8002/health

Docker Compose integration: The docker-compose.yml at the repository root provisions the shared infrastructure (PostgreSQL, Qdrant, Redis, MinIO) for the full AuraLearn platform. The llm-tutor-backend runs as an independent service alongside the summarizer backend, reminder system, and frontend.